

ECON 707/807: Econometrics II

8-Week Accelerated Session — Asynchronous Online

Department of Economics
Strome College of Business
Old Dominion University

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Course Format:	Fully online, asynchronous (Canvas)
Virtual Office Hours:	Thursday and Friday, 2:30 pm – 4:00 pm ET, via Zoom (in person by appointment)

1 Course Description

The objective of this course is to introduce students to time series analysis and forecasting. The goal is for students to obtain hands-on experience using financial, economic, business, crime, traffic, and other data to construct and evaluate forecasts. The course introduces several techniques for applied time series analysis and forecasting. Students will use R to enhance their computer skills.

This is an accelerated eight-week section covering the same material as the standard sixteen-week course. Expect roughly twice the weekly workload of a full-semester course: plan on approximately 15–18 hours per week for reading, recorded lectures, R work, assignments, and the forecasting project. Because the pace is compressed, falling one week behind is difficult to recover from.

1.1 Course Format and Weekly Rhythm

There are no live class meetings. The course is organized into eight Canvas modules, one per week. Each module opens at the start of its week and contains the assigned reading, recorded lecture videos, annotated R scripts, and the graded work for that week.

Unless stated otherwise, all module work is due at **11:59 pm ET on the Sunday that closes the module week**. Modules remain open for review after their due dates, so you may work ahead, but you may not fall behind.

1.2 Use of Statistical Software — R and R Studio

As this course focuses on the learning and application of time series techniques, a computer or laptop with R and R Studio installed is required for module work, assignments, and exams. Students can obtain the latest version of R from <https://www.r-project.org/> and a free version of R Studio from <https://www.rstudio.com/>. While students can code directly in R, it is recommended that students use R Studio to facilitate interactions with the R language. Students will also install numerous open source packages that facilitate economic analysis in R.

You may use Posit Cloud (formerly R Studio Cloud) instead of a locally installed version of R. You can create a free account at <https://posit.cloud/>.

1.3 Technology Requirements

- Reliable internet access and a computer capable of running R and R Studio.
- Access to Canvas, where all content, submissions, and grades are handled.
- Ability to render R Markdown or Quarto documents to PDF or HTML for submission.
- A backup plan for internet or hardware failure. Technology problems are not grounds for an extension in an eight-week course; submit early.

1.4 Prerequisites/Corequisites

Prerequisite: ECON 706 or equivalent.

2 Course Objectives

At the end of this course, students will be able to:

1. Understand and apply common statistical methods used in business and economic forecasting
2. Explain, identify, and extract components of a time series
3. Collect, organize, visualize, and analyze time series data
4. Interpret and critique forecasts

3 Textbooks and Other Materials

3.1 Required Texts

None. All required readings are freely available online and linked in Canvas.

3.2 Recommended Texts

- Francis X. Diebold, *Forecasting in Economics, Business, Finance and Beyond* — <https://www.sas.upenn.edu/~fdiebold/Teaching221/Forecasting.pdf>
- Hyndman, R.J., and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd edition. OTexts: Melbourne, Australia — <https://otexts.com/fpp3/>
- Wooldridge, Jeffrey M. *Introductory Econometrics: A Modern Approach*, 7th ed. (ISBN 9781337558860)

3.3 Reference Texts

- Gonzalez-Rivera, Gloria (2012). *Forecasting for Economics and Business*. Pearson.
- Hansen, Bruce — <https://www.ssc.wisc.edu/~bhansen/390/2010/>
- *A Little Book of R for Time Series* — <https://a-little-book-of-r-for-time-series.readthedocs.io/en/latest/src/timeseries.html>

4 Instructional Methodologies

This course is based on recorded lecture and practical application. Each student is expected to complete the assigned readings and recorded lectures before attempting that module's graded work. Each module includes worked R examples that you should run and modify yourself; watching the code is not a substitute for writing it. Students are expected to complete homework to apply the concepts covered in each module.

5 Requirements and Grading

The evaluation for this course consists of homework assignments, two exams (including a cumulative final examination), and a forecasting project. Due to the accelerated nature of the course, late submissions of any graded material will receive a score of zero once the answer key is posted. The final grade is the sum of the following elements.

Assignment	Points	Due (11:59 pm ET)
HW 1	50	End of Week 1
HW 2	50	End of Week 2
HW 3	50	End of Week 3
HW 4	50	End of Week 5
HW 5	50	End of Week 6
Project Pitch	50	End of Week 3
Project Data	50	End of Week 5
Final Project	200	End of Week 7
Midterm Exam	200	During Week 4
Final Exam	250	During Week 8
Total	1000	

5.1 Forecasting Project

The goal is for everyone to design an individual project, make decisions, and have some fun. You will choose a specific time series, make a set of forecasts, and evaluate your success. You will work on this project throughout the session, submitting a pitch, a data report, and a final project report. The forecasting project consists of three deliverables.

You will select a time series to forecast. It must have at least one new realization between when you submit your forecast and when you submit your evaluation. For example, if you want to forecast the U.S. unemployment rate for a given month, the number is announced by the BLS early in the following month; your forecast must be submitted before that release. If you want to forecast quarterly U.S. GDP, it is announced by the BEA at the end of the month following the quarter; your forecast must be submitted before that release.

Because the session is only eight weeks long, series selection matters more than it does in a full semester. Your pitch must identify the release date of the realization you intend to forecast, and that date must fall before the end of Week 7. Series whose next release falls outside the session will not be approved.

5.2 Exams

There are two exams. The midterm exam covers Modules 1–3. The final exam is comprehensive and covers Modules 1–7. Both are administered in Canvas as timed, single-attempt exams during a defined window, and both mix conceptual, theoretical, and interpretive questions. You may use R and your own notes; you may not consult another person or an AI system.

5.3 Homework

There are five homework assignments, submitted through Canvas. Each requires you to produce and interpret output in R. Submit both your write-up and your code.

5.4 Grading Scale

Grades will be converted using the following table.

Min	Max	Letter
930	1000	A
900	929	A–
870	899	B+
830	869	B
800	829	B–
770	799	C+
730	769	C
700	729	C–
670	699	D+
630	669	D
600	629	D–
0	599	F

Except for grades of “Incomplete,” all grades are considered final when reported by a faculty member at the end of a term. A change in grade may only be requested when a calculation, clerical, administrative, or recording error is discovered in the original assignment of a course grade, or when a decision is made by the faculty member to change the course grade because of disputed academic evaluation procedures.

Grade changes necessitated by a calculation, administrative, or recording error must be reported within a period of six months from the time the grade is awarded. No grade may be changed as the result of a re-evaluation of a student’s work or the submission of supplemental work following the close of a term.

6 Participation and Engagement

Per university policy, students are expected to participate in class. In an asynchronous course, participation means completing each module’s activities within its week. Canvas records module progress and submission times, and these records are used to document participation.

Students who fail to submit graded work for more than one module week are not making satisfactory progress and are subject to failure. If circumstances put you behind, contact me immediately rather than after the fact; in an eight-week course there is very little room to recover.

7 Online Conduct and Expectations

All course communication — email, Canvas messages, and Zoom office hours — should follow the same standards of professionalism expected in a workplace. When attending virtual office hours, join from a quiet location and be prepared with your specific question, your code, and the error message you are seeing. The more information you can provide, the more useful my response will be.

8 Course Disclaimer

The course schedule and activities are subject to change. Changes will be posted as Announcements in Canvas.

9 Cultural Diversity

Old Dominion University fosters a campus community that values and supports the cultural identities of each of our members. The University also fosters an inclusive environment and provides programs that cultivate a climate of awareness, understanding, and respect of diverse individuals and groups. [Source: Adapted from The Office of Intercultural Relations]

10 Honor Pledge

“I pledge to support the honor system of Old Dominion University. I will refrain from any form of academic dishonesty or deception, such as cheating or plagiarism. I am aware that as a member of the academic community, it is my responsibility to turn in all suspected violators of the honor system. I will report to Honor Council hearings if summoned.” By attending Old Dominion University, you have accepted the responsibility to abide by this code. This is an institutional policy approved by the Board of Visitors.

11 Academic Integrity

The Office of Student Conduct & Academic Integrity (OSCAI) oversees the administration of the student conduct system, as outlined in the Code of Student Conduct. Through their interactions with students, they attempt to foster a climate of personal and academic integrity that facilitates the success of all University community members. For more information, please visit the Honor Council online at The Office of Student Conduct and Academic Integrity.

Cheating: Copying, collaborating, or completing an academic exercise with or for other students or parties without permission from the class-assigned faculty member; paying someone to complete assignments; using any unauthorized materials to assist on assignments; misusing study aids such as Chegg, Quizlet, Course Hero, etc., to complete or pass class assignments; using test banks or copying answers from another source or student. This also includes **unauthorized use of artificial intelligence sites, including, but not limited to, ChatGPT, Dall-E, Alpha Code, Tensor Flow, Scribe, etc., and any act or behavior that gives the student an unfair advantage.** Please use AI as a resource to learning and not a substitute to your submission. AI use is not permitted on either exam.

12 Academic Conduct

Academic dishonesty will not be tolerated. In addition to imposing penalties ranging anywhere from a failing grade (F) to suspension, it is University policy to record all Academic Dishonesty convictions permanently on the student's transcript. I report all cases of academic dishonesty to the Office of Student Conduct.

13 Educational Accessibility and Accommodations

Old Dominion University is committed to ensuring equal access to all qualified students with disabilities in accordance with the Americans with Disabilities Act. The Office of Educational Accessibility (OEA) is the campus office that works with students who have disabilities to provide and/or arrange reasonable accommodations.

If you experience a disability that will impact your ability to access any aspect of my class, please present me with an accommodation letter from OEA so that we can work together to ensure that appropriate accommodations are available to you. Because the session is compressed, please do this in Week 1 so accommodations can be in place before the midterm.

If you feel that you will experience barriers to your ability to learn or take tests in my class but do not have an accommodation letter, please consider scheduling an appointment with OEA to determine if academic accommodations are necessary.

The Office of Educational Accessibility is located at 1021 Student Success Center, and the phone number is 757-683-4655. Additional information is available on the OEA website: The Office of Educational Accessibility.

14 Communication

Students should feel welcome to contact me via email (rsitti@odu.edu) or join my virtual office hours. I respond to well-crafted emails within 24 business hours. Students should take the time to craft complete, professional emails. The more information that you can provide about a question or problem, the more likely that my response will be helpful. Avoid non-professional language and practice communicating in the corporate workplace. Emails that are unprofessional will be returned with no action. There are many guides on how to compose a professional email which you can easily find online.

15 University Email Policy

The Old Dominion University email system is the official electronic mail system for distributing course-related communications, policies, announcements, and other information. In addition, the University email user ID and password are necessary for authentication and access to numerous electronic resources (online courses, faculty webpages, etc.). For more information about student email, please visit Student Computing.

16 Withdrawal

A syllabus constitutes an agreement between the student and the course instructor about course requirements. Participation in this course indicates your acceptance of its teaching focus, require-

ments, and policies. Please review the syllabus and the course requirements as soon as possible. If you believe that the nature of this course does not meet your interests, needs or expectations, if you are not prepared for the amount of work involved — or if you anticipate that the accelerated pace, assignment deadlines, or abiding by the course policies will constitute an unacceptable hardship for you — you should drop the class by the drop/add deadline, which is listed in the ODU Schedule of Classes. Note that in an eight-week session the add/drop and withdrawal deadlines arrive much sooner than in a full semester. For more information, please visit the Office of the University Registrar.

17 Other Academic Policies

Please see the following link for other academic policies at the university level: <https://catalog.odu.edu/undergraduate/academicinformation/>

18 Tentative Course Schedule

Wk	Module / Topic	Reading	Due
1	Visuals, data, and the forecasting workflow	Diebold Ch. 1–2; FPP3 Ch. 2	HW 1
2	Naïve and benchmark forecasts; forecast types and evaluation	Diebold Ch. 4–6	HW 2
3	Autocorrelation and stationarity	Diebold Ch. 7–8	HW 3; Project Pitch
4	AR and MA models; ARMA forecasting	Diebold Ch. 9–12	Midterm Exam (Modules 1–3)
5	Trend, seasonality, and time-series decomposition	Diebold Ch. 13–14	HW 4; Project Data
6	x variables, cointegration, and dynamic regression	Diebold Ch. 15–16; FPP3 Ch. 9	HW 5
7	Multivariate forecasting: VAR and impulse response functions	FPP3 Ch. 11	Final Project
8	Review and comprehensive final examination	—	Final Exam

Note. Spatial modeling, covered in the sixteen-week version of this course, is not part of the eight-week schedule. Materials for that topic will be posted in Canvas as optional enrichment for students who wish to pursue it in their forecasting project.